

AeroFlow AI

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PROBLEM STATEMENT



The "Stop-and-Go" Crisis

Vehicles idling in traffic or moving in stop-and-go waves emit up to 20× more pollutants than those driving smoothly. Urban traffic accounts for 30–50% of PM_{2.5} and NO_x emissions.

Current Intelligent Transport Systems (ITS) only react to congestion. They ignore environmental impact until a smog alert is already triggered - making urban management entirely reactive rather than preventive.



ABSTRACT



AeroFlow bridges the gap between Urban Traffic Management and Environmental Protection. Instead of reacting to pollution after it reaches critical levels, our Machine-Learning model predicts local PM2.5 spikes up to 6 hours in advance and dynamically adjusts traffic routing to prevent smog accumulation.



TECH STACK

- Frontend UI: Next.js, Tailwind CSS
- Mapping & Visualization: React-Leaflet (OpenStreetMap)
- Backend API: Python, FastAPI
- Machine Learning: Scikit-learn - Random Forest for PM2.5 prediction
- $R^2 = 0.97$ · MAE $\approx 5 \mu\text{g}/\text{m}^3$ · top predictors: traffic density (53%), road topology (35%), wind speed (10%)
- Generative AI: Groq API (Llama 3.3 70B) for automated reporting
- Deployment: Vercel (Frontend), Render (Backend)

EXISTING SOLUTIONS & DRAWBACKS



- Existing ITS (Intelligent Transport Systems): focus solely on vehicle throughput.
 - Drawback: ignores public health and air quality completely.
 - Environmental Dashboards: provide real-time data and alerts.
 - Drawback: strictly reactive, they inform citizens only after the air is already toxic.
 - Blanket Traffic Bans: often implemented during smog alerts.
 - Drawback: cause severe public frustration and economic loss; unnecessary if micro-routing is applied early.
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PROPOSED SOLUTION



- Proactive Prediction: a Random Forest model forecasts "smog islands" before they form, using traffic density, road topology, and wind data.
 - validated: the model's top learned features are traffic density, street-canyon topology, and wind confirming the physical hypothesis.
 - Dynamic Redirection: simulates diverting traffic away from critical zones (e.g. schools, hospitals). Round 2 integrates with existing SCADA / traffic-controller APIs to extend real "green waves" on perimeter roads.
 - GenAI Reporting: instantly generates compliance and action reports using an LLM, keeping stakeholders informed without manual data synthesis.
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ARCHITECTURE / WORKFLOW

1. Data Ingestion: collects traffic-density and weather data (mocked for MVP via a digital-twin simulator).
2. Prediction Engine: the Random Forest model predicts future PM2.5 per city sector.
3. Decision Logic: if predicted PM2.5 exceeds safe limits → trigger the Traffic Redistribution Algorithm. (MVP: simulated redistribution; production: SCADA hook.)
4. Output & Visualization: the dashboard updates heatmaps and shows the optimized scenario.
5. LLM Alerting: the Groq API generates a formal alert and mitigation summary.

UNIQUENESS / INNOVATION



Proactive vs Reactive: the first system designed to "cure" city lungs before they get sick.

Zero-Hardware Deployment: \$0 in new physical infrastructure, integrates with existing city-sensor and traffic-controller APIs.

Cross-Domain Synergy: merges two isolated domains (ecology and transportation) into one actionable AI model.

High Speed & Low Cost: a lightweight Random Forest ensures rapid response times.



ADVANTAGES & LIMITATIONS



Advantages:

- Direct reduction in respiratory diseases and healthcare costs.
- Highly scalable to any city with standard traffic-management APIs.
- User-friendly, high-contrast command-center UI for non-technical city officials.

Limitations:

- Relies on the accuracy of existing third-party weather and traffic sensors.
- Unpredictable events (e.g. sudden major accidents) can temporarily skew routing efficiency.

AeroFlow AI is a working coded prototype that includes ML prediction, live traffic optimization, and GenAI reporting:

All of these are already implemented; Round 2 focuses on real-sensor data and SCADA integration.



CONCLUSION



Smart cities shouldn't just be fast – they must be breathable.

AeroFlow AI transforms passive environmental data into active, life-saving infrastructure control. By anticipating pollution before it settles, we reclaim urban health without sacrificing mobility.

Impact · Feasibility · Innovation.



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